

# Computer Arithmetic & Digital Logic Questions

## Questions 1 to 26

1. Why computers use the binary number system?
2. Add binary numbers 1011 and 101 in both decimal and binary forms.
3. Add binary numbers 1010110 and 1011010.
4. Add binary numbers 10111 and 1011.
5. Find the complement of the following numbers:

(a)  $495_{10}$

(b)  $29_{10}$

(c)  $48_8$

(d)  $C_{16}$

(e)  $2_5$

(f)  $32_4$

6. Find the complement of the following binary numbers:

(a)  $10_2$

(b)  $101_2$

(c)  $101101_2$

(d)  $011011_2$

(e)  $10110001_2$

(f)  $001101001110_2$

7. Subtract  $011011_2$  from  $1101110_2$ .
8. Subtract  $0101_2$  from  $10000_2$ .
9. Subtract  $011011_2$  from  $110111_2$ .
10. Subtract  $25_{10}$  from  $50_{10}$  using complementary method.
11. Subtract  $25_{10}$  from  $20_{10}$  using complementary method.
12. Subtract  $234_{10}$  from  $588_{10}$  using complementary method.
13. Subtract  $216_{10}$  from  $172_{10}$  using complementary method.

- 14.** Subtract  $0101_2$  from  $10000_2$  using complementary method.
- 15.** Subtract  $110111_2$  from  $101110_2$  using complementary method.
- 16.** Subtract  $011011_2$  from  $110111_2$  using complementary method.
- 17.** Subtract  $1111_2$  from  $1100_2$  using complementary method.
- 18.** Multiply binary numbers 1100 and 1010.
- 19.** Multiply binary numbers 0101 and 1001.
- 20.** Multiply binary numbers 101111 and 111.
- 21.** Divide  $11001_2$  by  $101_2$ .
- 22.** Divide  $01010111_2$  by  $0111_2$ .
- 23.** What happens when a computer divides a number by zero?
- 24.** Explain how a computer performs multiplication and division operations internally by using additive approach.
- 25.** What is the primary advantage of performing subtraction by complementary method in digital computers?
- 26.** Discuss the advantages and disadvantages of performing various arithmetic operations by additive method in a digital computer.